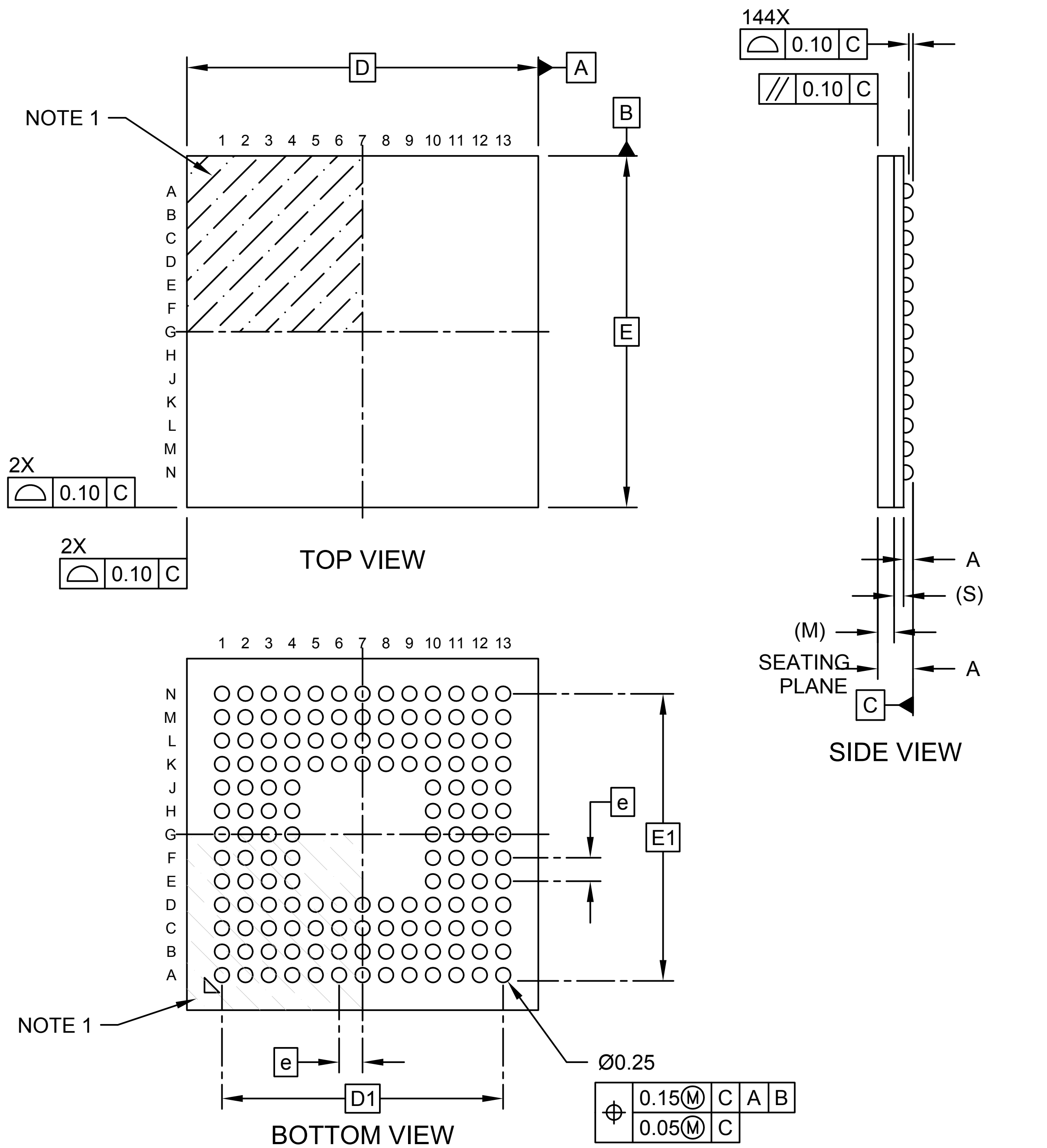
144-Ball Ultra Thin Fine Pitch Ball Grid Array (CVB) - 6x6x0.6 mm Body [UFBGA] Atmel Legacy Global Package Code CBH

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



144-Ball Ultra Thin Fine Pitch Ball Grid Array (CVB) - 6x6x0.6 mm Body [UFBGA]
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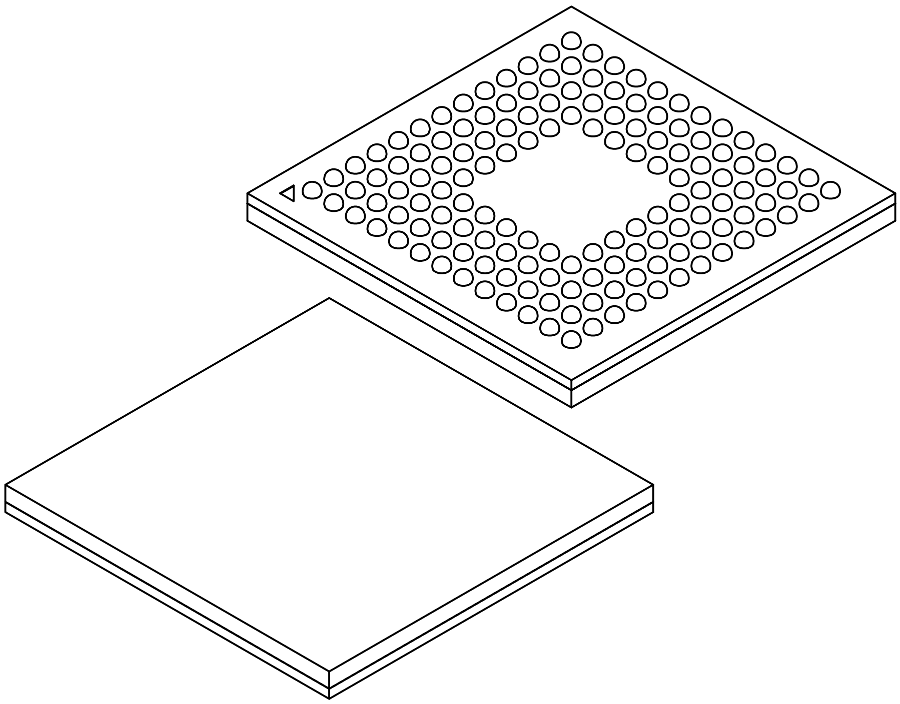

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		144	
Pitch	e		0.40 BSC	
Overall Height	A	–	–	0.60
Standoff	A1	0.11	0.16	0.21
Substrate Thickness	S		0.136 REF	
Mold Cap Thickness	M		0.25 REF	
Overall Length	D		6.00 BSC	
Overall Terminal Pitch	D1		4.80 BSC	
Overall Width	E		6.00 BSC	
Overall Terminal Pitch	E1		4.80 BSC	
Ball Diameter	b	0.22	0.25	0.28

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

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144-Ball Ultra Thin Fine Pitch Ball Grid Array (CVB) - 6x6x0.6 mm Body [UFBGA] Atmel Legacy Global Package Code CBH

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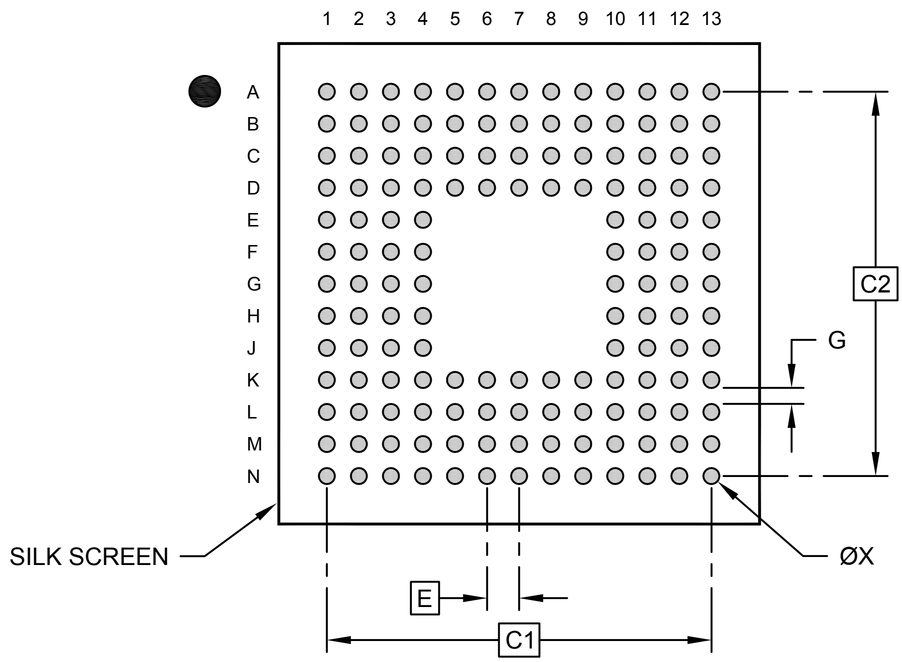

RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.40 BSC	
Contact Pad Overall Pitch	C1		4.80 BSC	
Contact Pad Overall Pitch	C2		4.80 BSC	
Contact Pad Diameter (X144)	X			0.20
Contact Pad to Contact Pad	G	0.20		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23475 Rev A